CHAPTER

12 Mathematical Methods for Economics

"The master economist must possess a rare combination of gifts. He must be mathematician, historian, statesman, philosopher to some degree"

- J.M.Keynes

Learning Objectives

1. To understand why mathematics is required for economics,
2. To learn the knowledge of mathematical methods, as a facility for self-expression not only in descriptive economics, but also in quantitative economics.

12.1 Introduction

Economic analysis is a systematic approach to (a) determine the optimum use of scarce resources and (b)choose available alternatives and select the best alternative to achieve a particular objective. Mathematical methods are helpful for achieving the objectives of the economic analysis.

12.1.1 Why Study Mathematics?

The subject Economics deals with many quantitative variables and functions, in consumption, production, distribution and policy making. Hence, the mathematical methods would help economists to use the quantitative variables in a better way and to obtain accurate results.

The lengthy and descriptive economic contents can be clearly set in simple notation in mathematical models for clear and easy understanding. For example, **the number of pens demanded in a given time period in a Higher Secondary School is 200 when price is zero. This decreases by 10 for every ₹1 rise in the price of pen.** It is expressed mathematically as

Q = 200 – 10 P, here Q is the quantity demanded and P is the price. Thus large information can be expressed and communicated with simple functions and equations.

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